

Lab 2: LINEAR REGRESSION

DUE: 7:50am 16/04/2025

Submission: mlearning

- **Lab2_YOURNAME.ipynb**
- Must include all data file (.csv,...) and .ipynb in your submission
- .csv and .ipynb must be in the same folder (pd.read_csv('data.csv'))
- No use of LOCAL libraries/folders
- NO LATE submission accepted

For reference, please read **Building a single-variable regressor**, Page 117 in the Textbook, Apply Linear Regression to predict house prices for each of the following cases.

For each case:

- normalize data using MinMaxScaler()
- use 80% data for training and 20% for testing.
- compute **MSE** score for testing data
- compute the **coefficient of determination R2 score** for testing data for each case

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

- write the Linear Equation for predicting house prices

There are 6 cases:

1. Use only area
2. Use only area, bedrooms
3. Use only area, bedrooms, bathrooms
4. Use only area, bedrooms, bathrooms, stories
5. Use only area, bedrooms, bathrooms, stories, parking
6. Use only area, bedrooms, bathrooms, stories, parking, airconditioning (yes->1, no->0)

Summarize MSE, R2 metrics of 6 cases in a Table